UI / UX AUDIT · REVISION 2

# Molepipe

Material 3 design language & features,
and consistency across panels, tabs, buttons, navigation & selection

**Focus:** Material 3 conformance (colour · type · shape · elevation · motion · state layers · components) and one-grammar consistency across every surface — tabs and panels, button hierarchy, ways of navigating, ways of selecting
**Scope:** full application UI — onboarding wizard, lock/security, messaging, media viewer, settings, wide/two-pane layouts
**Evidence:** `ui-audit/` capture set — 68 portrait + 49 landscape + 7 retest screenshots (Android 16 emulator, 1080×2400) with a scripted walkthrough README; byte-level pixel analysis of the viewer/landscape sets; line-level re-audit of the shipped WebView source (`backend/internal/api/public/{index.html, style.css, app.js, material-theme.css, vendor/}`)
**Source of truth:** every finding is verified against code locations (`file:line`) or pixel measurements; the walkthrough README is treated as claims to verify
**Date:** October 2026 · **Build audited:** debug `org.molepipe.molepipe` (commit `da60c88-dirty` era)
**Status:** revision complete — M3 polish pass (Rev 1) implemented and deployed; this revision re-scopes the report onto design language and cross-surface consistency, and adds a second consistency pass over the current source (16 open gaps, §12). Screenshot re-capture pending

## Contents


1. Scope & method
2. Executive summary — M3 scorecard & consistency matrix
3. Material 3 foundations — colour, type, shape, elevation, motion
4. Material 3 components & features
5. Consistency · panels & tabs
6. Consistency · buttons & action hierarchy
7. Consistency · navigation
8. Consistency · selection
9. Flow-level findings (preserved from Rev 1)
10. Strengths preserved
11. M3 polish pass — what changed
12. Open gaps & M3 migration roadmap
13. Verification
14. Appendix A — screenshot index
15. Appendix B — finding ID map (Rev 1 → Rev 2)


## 1 · Scope & method

The `ui-audit/` folder documents a complete user journey captured over CDP-driven interaction and `adb screencap`: fresh install → account creation → friend request → two-way messaging (text + images) → lock/security states, in both portrait and a landscape pass that crosses the app's 900 px two-pane breakpoint, plus an image-send retest.

Revision 2 narrows the lens to two questions:

- **Material 3 design language & features** — are the M3 colour roles, type roles, shape scale, elevation ladder, motion and state layers actually driving the UI, and which M3 features/components are used, hand-rolled, or vendored-and-ignored?
- **Consistency across surfaces** — when the same job appears on different panels and tabs (a primary action, a cancel, a back, a selection, a selected state), does it look, read and behave the same way, all the way down to individual buttons and to how people navigate and select?

Findings were produced by cross-referencing three evidence streams:

- **Walkthrough claims** — the state descriptions in `ui-audit/README.md`;
- **Pixel analysis** — per-frame luminance/colour sampling, region diffs and MD5 de-duplication of the capture sets (which corrected several README captions, e.g. `54-image-delivered.png` is byte-identical to the failed state, and the "actions bar" frames are duplicates);
- **Source audit** — line-level review of the four UI files and the vendored Material Web bundle (a WebView shell serving one HTML/CSS/JS app with generated Material 3 colour tokens).

Severities: High functional breakage, severe UX failure, or accessibility blocker · Medium consistency/hierarchy defect visible in normal use · Low polish. Status: Fixed · Partial · Open (this consistency pass) · Backend. Findings carried over from Rev 1 keep their legacy ID as (was X-n).

## 2 · Executive summary

The app's Material 3 *foundations* are now genuinely strong: a complete 36-role colour system with seven accents plus runtime Material You palettes, a documented shape and type scale, correct M3 elevation recipes, and honest Tor/destructive-confirmation UX. The remaining risk is not the foundations — it is **drift between surfaces**. The same concept still resolves differently depending on where you stand:

- **Three switch mechanics for three tiers of navigation** (bottom nav, wizard, You sub-tabs) with three different active-state and ARIA expressions;
- **Selected-state colouring is split** — segmented toggles highlight on `primary-container` while tabs, theme buttons and nav highlight on `secondary-container`;
- **Button hierarchy varies per surface** — the confirm dialog is the only dialog whose Cancel is a filled-tonal twin of its confirm; "Copy", "Scan QR" and "Cancel" each appear in two or three different button styles; Accept/Decline changes grammar per request type;
- **Navigation and selection affordances are close but not one system** — back/close glyphs fork three ways; multi-select carries no ARIA at all while the You tabs carry a half-implemented one.

Underneath that drift, the Material 3 *component* layer is only half-adopted: buttons, text fields, switch, list and linear progress are real `md-*` elements, but dialogs, menus, the navigation bar, tabs and the FAB are hand-rolled against the M3 spec — while `md-dialog`, `md-menu`, `md-tabs`, `md-ripple` and `md-focus-ring` sit vendored and unused (§4, §12).

| | Count | Fixed | Partial | Open (this pass) | Backend |
|---|---|---|---|---|---|
| High | 13 | 13 | 0 | 0 | 0 |
| Medium | 27 | 10 | 4 | 12 | 1 |
| Low | 4 | 0 | 0 | 4 | 0 |
| **Total** | **44** | **23** | **4** | **16** | **1** |

## Material 3 conformance scorecard

| M3 area | State | Notes |
|---|---|---|
| **Colour roles** | Strong | Full 36-role system set + surface ladder (`material-theme.css`); 7 accent schemes; runtime Material You palettes via `AndroidShell.systemPalette()` (`app.js:207-241`). ~20 hex literals remain in scrims/viewer/QR paths; `.danger-text` hardcodes error hexes (M3-9) |
| **Tonal surfaces / elevation** | Strong | Cards, nav, menus, modal, sheets drawn from the `surface-container-*` ladder; M3 shadow recipes in `--elev-1…3` |
| **Shape scale** | Strong | Role-per-radius `--shape-xs…full` used ~63× (M3 Expressive mapping); literals only on bubble tails and one documented cut-corner |
| **Type roles** | Good | 8-role scale mapping to M3 title/body/label roles; 63/69 font-size rules tokenised; 6 raw sizes + dead `--type-mono` remain (M3-6) |
| **Motion** | Weak | `--ease-emphasized` + `--dur-fast/med/slow` defined and used **0 times**; animations run on custom springs and one-off easings (M3-5). `prefers-reduced-motion` coverage is thorough |
| **State layers / ripple** | Weak | Hand-rolled controls fake state layers with `color-mix` hovers and a global `scale(0.93)` press; `md-ripple` and `md-focus-ring` are vendored but never used (M3-13) |
| **Components** | Mixed | Real `md-*`: filled/tonal/text buttons, filled/outlined text fields, switch, list, linear progress. Hand-rolled: dialogs, menus, nav bar, tabs, FAB, keypad, swatches, select bar (M3-12) |
| **Targets & focus** | Good | `--tap-min: 48px` honoured on icon buttons/segments/nav; several controls still 44px and two strips have no min-height (BT-5); global focus ring present |

## Cross-surface consistency matrix

How the same job is expressed on each surface. Divergent cells are the subject of §5–§8.

| Surface | Primary action | Secondary / cancel | Back / dismiss | Selection | Selected / active state |
|---|---|---|---|---|---|
| Onboarding wizard | `md-filled` "Next" (footer, right) | `md-text` "Back" (footer, left); in-panel primaries | Footer Back + hardware chain | — | Step dots (done/active) + "Step N of M" `aria-live` |
| Chats (empty) | `md-filled-tonal` "Add someone" | `md-text` "Scan QR", "Create group" | — | — | Nav pill indicator (class only) |
| Chats (list) | Extended FAB "New chat" | Row tap → detail | `#back-btn` `ic-back` arrow | Long-press → menu; select mode | Nav pill indicator |
| Requests | **Two** `md-filled-tonal` peers (Accept / Decline) | Relay & group invites use `md-text` / `md-text` | — | — | Nav pill indicator |
| You › Profile | `md-filled` "Copy" | `md-filled-tonal` "QR code"; `md-text` "New code" | Hardware back → Chats | Accent swatch (ring + check) | `.you-tab` pill (`secondary-container`) |
| You › Security | `md-filled` saves (duress/wipe wear `.danger-fill`) | `md-text` cancels; hand-rolled `.danger-outline` pill | Hardware back → Chats | `.seg` wipe threshold | `.seg.active` = `primary-container` (!) |
| Add screen | `md-filled` "Look up" / "Send request" | `md-filled-tonal` "Scan QR", "Paste" | `#back-btn` arrow | `.seg` person / group | `.seg.active` = `primary-container` (!) |
| Chat detail | `.send-btn` 44px circle | Attach (icon), overflow menu | Arrow (`ic-back`); forced hidden ≥900px | Long-press menu; select mode | `.selected` 8% primary tint |
| Image viewer | Icon actions (download/share/forward) | — | `ic-close` × + backdrop tap | — | Zoom % pill on gesture end |
| Dialogs (edit family) | `md-filled` confirm (right) | `md-text` Cancel (left) | Esc / hardware back | — | Focus trap + restore |
| Dialog (confirm) | `md-filled-tonal` confirm (right) | **`md-filled-tonal` Cancel (left)** (!) | Esc / hardware back | — | Focus trap + restore |
| Lock screen | `md-filled` "Unlock" | `md-text` "Forgot PIN" | — | Keypad keys | PIN dots |
| Select mode | Icon actions in `#select-bar` | `#select-close` | Hardware back exits | `.select-check` + tint | "N selected" contextual bar |

## 3 · Material 3 foundations — colour, type, shape, elevation, motion

**M3-1** Three live token bugs **High** **Fixed** Tokens (was DS-1)

`--elev-3` used but never defined → the long-press message menu rendered with *no shadow*; `@keyframes menu-pop` referenced but never defined → reaction bar had no entrance animation; `--shape-md` typo (real name `--shape-m`) masked by a fallback in two notice banners.

style.css:1157 · style.css:1231 · style.css:742, 755

**Resolution:** token defined, animation renamed to `popover-in`, typos corrected.

**M3-2** No type, spacing, elevation or z-index tokens — 22 ad-hoc font sizes **High** **Fixed** Tokens (was DS-2)

Colour was ~85% tokenised, but every margin/padding/gap and font-size was a hand-picked literal: 22 distinct sizes (0.66–1.6rem + one `13px`), ~25 padding steps, 9 scrim opacities, 11 shadow recipes, 5 literal `999px` pills, 15 ad-hoc z-index values (including a dead `1000` that never applied). Near-duplicate caption sizes (0.66/0.68/0.70/0.72/0.75/0.78) were noise, not design.

style.css:92–96 (5 documented roles vs 22 used) · throughout

**Resolution:** full token layer: `--type-*` (8 roles mapped to M3 type roles), `--space-1…6`, `--elev-1…3` + `--elev-sheet`, `--scrim`/`--scrim-heavy`, motion vars, layout constants (`--nav-height`, `--tap-min`, `--content-max`), 8-step z-index scale, semantic success/warning colours. 114 sites codemodded onto tokens.

**M3-3** Hardcoded colour islands breaking theming **Medium** **Fixed** Colour roles (was DS-4)

The donate card hard-coded a five-hex Material-pink palette with no dark-mode path (its muted body text fell to ≈1.4:1 contrast in dark); presence/strength used two greens and three ambers outside the token system; 34 inline `style=` attributes in HTML, including an entire medallion component.

style.css:281–293 · :304, 722–723, 971–972 · index.html (34 inline styles)

**Resolution:** donate card rebuilt on `tertiary-container` tokens (theme-adaptive, AA in both schemes); status colours consolidated to `--color-success`/`--color-warning`; inline styles reduced 34 → 4 (remaining are legitimate layout one-offs).

**M3-4** Duplicate selectors silently overriding each other **Medium** **Fixed** Tokens (was DS-3)

`.modal-backdrop` and `.modal-card` each defined twice (the later block silently won, leaving a dead `z-index:1000` and duplicate shadow recipes); `.row-between` defined twice with different gaps (tuning one screen unknowingly changed chat-list rows app-wide); `#add-screen` and `.pill-label` also duplicated.

style.css:356/485 · :138/:803 · :597/:847 · :1328/:1395

**Resolution:** merged into single definitions; dialogs given one M3-spec block at `--z-dialog` (above snackbars).

### M3-5 M3 motion tokens exist but are never used

Medium · Open · Motion

`--ease-emphasized: cubic-bezier(0.2, 0, 0, 1)` and `--dur-fast/med/slow` are defined alongside the M3-emphasized curve — and referenced **zero** times across `style.css`, `app.js` and `index.html`. Actual motion runs on custom springs (`--spring-emph`, `--spring-press`) and one-off `ease`/ms literals (e.g. wizard dots 0.2s, toast 220ms, menu 180ms). The result is a motion language that is neither M3 nor documented — every surface animates slightly differently.

style.css:46–54 (defined) · :297, 634, 875, 1273 (one-offs) · reduced-motion coverage :1775–1787

**Open gap:** pick one policy — either adopt `--ease-emphasized` + durations for surface transitions (M3) or delete the dead tokens and document the spring system; then route dots/toast/menu/viewer through it.

### M3-6 Spacing scale unadopted; dead tokens linger

Low · Open · Tokens

`--space-1…6` is used exactly once (`--space-5` in the modal card); ~208 padding/margin/gap declarations still carry raw px/rem literals. `--type-mono` and `--weight-bold` have zero uses, `--dur-*` zero uses (M3-5), and 6 raw font sizes remain (style.css:512, 1090, 1348, 1352, 1357, 1620).

style.css:33, 36–44 · :554 (only `--space-5` use)

**Open gap:** either codemod spacing onto `--space-*` where values align to the 4px scale (the comment already concedes "optical sub-4px tweaks stay as literals") or remove the unused tokens so the scale means what it says.

### M3-7 Step headings barely larger than body text

Medium · Fixed · Type roles (was T-1)

`.title` was 1rem/500 over 0.95rem/400 body — a 0.8px difference — giving wizards and modals a flat, undifferentiated focal order; the confirm-phrase modal didn't use the app's own 1.25rem dialog-title role.

style.css:92–95

**Resolution:** type roles with real contrast: headline/title-large at 22px for pane headers & dialog titles, title at 16px, body 15px; empty-state titles aligned to the headline role at weight 500 (M3).

### M3-8 Password requirements failed AA contrast at their critical size

Medium · Fixed · Colour roles (was T-3)

The rules list telling users exactly what their password must satisfy used `--outline` at 0.78rem (≈3.9:1); the privacy footer was worse at 0.72rem.

style.css:372–375, 424–428

**Resolution:** muted text re-sourced onto `on-surface-variant` roles (≈7.6:1) with the caption/label type roles.

**M3-9** Error colour bypasses the error role in one class **Low** **Open** Colour roles

`.danger-text` hardcodes `#ba1a1a`/`#ffb4ab` instead of `var(--md-sys-color-error)`, so it will drift if an accent scheme or future palette ever moves error. Success/warning tokens are also outside the M3 role set — but those are documented as deliberate app-level semantic colours.

style.css:1125–1128 · :70–71 (documented)

**Open gap:** re-source `.danger-text` onto `--md-sys-color-error`/`--md-sys-color-on-error-container` roles.

## 4 · Material 3 components & features

**M3-10** Three modal dialects; 4 of 7 dialogs lacked roles; no focus management **High** **Fixed**

Components · dialogs (was C-2)

Div-modals, one native `<dialog>`, and divergent button grammars (Cancel as text-button vs tonal; generic "Save" on a security action). `#edit-dialog` had no `role`/`aria-modal` while `#confirm-modal` did; three title styles; two inline `max-width:420px` overrides; no focus trap anywhere; only two dialogs moved initial focus.

index.html:279, 531, 574, 863, 880, 892, 971, 982 · app.js:299–392

**Resolution:** one dialog spec (M3: surface-container-high, 28px corners, 24px padding, title-large headline), roles/aria on every dialog, shared Tab focus trap + focus-restore-to-trigger, action-specific confirm verbs ("Turn on PIN", "Erase everything"), `confirmLabel` support in `editDialog`. Full `md-dialog` migration deferred (§12).

**M3-11** Seven feedback channels; the snackbar had one caller **Medium** **Fixed**

Components · snackbar (was C-1)

A styled, `role="status"` snackbar existed but was used exactly once (friend accepts). Elsewhere: a red `role="alert"` paragraph carried **success** copy ("Relay request sent…"), four separate ad-hoc "Copied!" label swaps, a bespoke 12-second accept banner, and a 2-second auto-closing duress message. Toasts stacked at identical coordinates rather than queueing.

app.js:1210 (only caller) · app.js:2118 (success in alert sink) · app.js:1994, 2770, 3309, 2008

**Resolution:** M3 snackbar: one visible at a time (newest replaces), action-button support, M3 colour spec (inverse-surface / inverse-on-surface / inverse-primary action, `--shape-xs`); all copy confirmations, relay success and neutral notices migrated; success can no longer land in an alert region.

### M3-12 Half the Material Web bundle is vendored and unused; core chrome is hand-rolled

Medium · Open · Components

The bundle registers the full Material Web set, but the app only consumes buttons, text fields, switch, list and linear progress. Everything that gives M3 its interaction language — **dialogs, menus, tabs, icon buttons, FAB, ripple, focus ring** — is hand-rolled to look right rather than behave right. That is the root cause of the selection/navigation gaps in §7–§8: the hand-rolled menu cannot get keyboard interaction for free, and the hand-rolled tab strip cannot get roving tabindex for free.

| Used (`md-*`) | Vendored but unused | Hand-rolled equivalent |
|---|---|---|
| `md-filled-button`, `md-filled-tonal-button`, `md-text-button` | `md-elevated-button`, `md-outlined-button`, all `md-icon-button` variants | `.icon-btn`, `.send-btn`, `.fab`, `.seg`, `.mode-btn`, `.keypad-btn` |
| `md-filled-text-field`, `md-outlined-text-field` | `md-filled-select`, `md-outlined-select`, `md-slider` | — |
| `md-switch` | `md-checkbox`, `md-radio` | `.select-check` circle, relay picker radios |
| `md-list`, `md-list-item` (JS-created rows) | `md-divider`, `md-chip-*` | list rows, pills, badges |
| `md-linear-progress` | `md-circular-progress` | `.pill-spinner` SVG, `.wiz-dots` |
| — | `md-dialog`, `md-menu`/`md-menu-item`, `md-tabs`/`md-primary-tab`, `md-fab`, `md-ripple`, `md-focus-ring`, `md-elevation` | `.modal-backdrop`, `#msg-menu`/`.menu`, `.you-tabs`/`.navitem`, `.fab`, `color-mix` state layers, `:focus-visible` ring |

vendor/material-web.js (full bundle) · index.html + app.js tag census · app.js:451 (comment: "md-dialog would give this for free")

**Open gap:** the migration order in §12 (dialogs → menus → tabs) is what converts "looks like M3" into "behaves like M3".

### M3-13 State layers are faked; ripple and focus ring unused

Low · Open · State layers

Hand-rolled controls approximate M3 state layers with 8–10% `color-mix` hovers and `:active` fills, plus a global `scale(0.93)` press spring on every button type (which is expressive, but is not an M3 state layer and fights `md-*` components' own ripple). `md-ripple` and `md-focus-ring` are in the bundle and never used.

style.css:271–279 (press spring) · :236, 1021, 1279, 1350 (fakes) · :1768–1773 (global focus ring)

**Open gap:** adopt `md-ripple`/`md-focus-ring` on hand-rolled controls, or define `--state-hover/pressed` tokens and apply them uniformly; decide whether the press-scale belongs to a reduced-motion-aware motion system (M3-5).

## 5 · Consistency · panels & tabs

Three tiers of "switchable surface" exist — bottom-nav tabs (Chats/Requests/You), wizard steps, and the You sub-tabs (Profile/Preferences/Devices/Security) — plus detail panes (add screen, group info, chat) and in-modal micro-wizards (duress). The look converges; the *mechanics* do not.

### PT-1 Three different switch mechanics for three tiers of navigation

Medium · Open

Panels · mechanics

Bottom nav, detail panes and wizard steps swap via the `hidden` attribute (`setTab`, `showDetail`, `wizGoTo`); You sub-panes swap via a `.pane-on` class on `[data-you-pane]` wrappers; the duress setup is a third micro-pattern (`showDuressStep` toggling `hidden` on 3 in-modal steps with its own Back buttons). Anyone changing "how panes appear" must know which mechanism each tier uses.

app.js:746–768 (setTab) · :701–725 (showDetail) · :770–779 (showYouPane) · :3116–3143 (wizGoTo) · :2623–2625 (showDuressStep) · style.css:1707–1708

**Open gap:** one `showPane(el)` primitive (hidden + focus + optional history) for all tiers; You panes currently need the class form only because sibling wrappers share a pane id — that can become a single container per pane.

### PT-2 Active state and ARIA expressed three ways; none is a complete tab pattern

Medium

Open · Panels · ARIA

Bottom nav items: `.active` class only — no `aria-current`, no `aria-selected` (index.html:728–730). You tabs: `aria-selected` on `role="tab"` buttons inside `role="tablist"` — but no `aria-controls` anywhere in the app, no `role="tabpanel"` on the panes, no roving `tabindex`, no arrow-key handling. Wizard: no tab roles at all, but compensates with focus-to-heading + "Step N of M" `aria-live`. Three different announcements for "where am I".

app.js:762 (class-only) · :775 (aria-selected) · :3133–3142 (wizard focus) · index.html:408–413 (tablist without panels)

**Open gap:** decide the pattern per tier and finish it — M3 navigation bar semantics (`aria-current="page"`) for `.navitem`; full ARIA tabs (controls + roving tabindex + ←/→) for `.you-tabs`; wizard already models a linear flow correctly.

**PT-3** Selected-state colour and shape split across tab-like controls **Medium** **Open**

Panels · selection colour

All four "pick one" controls look like siblings but highlight differently: `.seg` (person/group, group policy, wipe threshold) uses **`primary-container`** on a `shape-m` track; `.mode-btn` (theme Auto/Light/Dark) uses **`secondary-container`** with no shared track at all (3 independent outlined pills); `.you-tab` uses `secondary-container` on a `shape-full` track; `.navitem` uses a `secondary-container` pill indicator. In M3 terms, primary-container means "selected" in segmented buttons and secondary-container in tabs — the app uses both for the same job with no rule.

style.css:1032–1043 (.seg) · :866–873 (.mode-btn) · :1683–1705 (.you-tab) · :727–735 (.navitem)

**Open gap:** write the rule down and follow it — recommended: **segmented buttons = primary-container, tab strips / nav = secondary-container** — then move `.mode-btn` into the `.seg` track pattern (it is a segmented control without a segment).

**PT-4** Empty states competed with the FAB; sibling CTAs mismatched **Medium** **Partial**

Panels · hierarchy (was C-4)

The Chats empty state stacked four actions while the "New chat" FAB floated simultaneously; the Requests empty state mixed tonal + outlined siblings. After the polish pass Chats is one tonal primary + two text secondaries and the FAB hides — but **Requests still presents two `md-filled-tonal` peers** ("Add someone", "Share my code") with no primary, and the detail/search empty states have no CTA at all.

index.html:376–381, 400–401, 751–755, 935–939 · app.js:675

**Resolved:** Chats empty state one tonal primary + text secondaries; FAB hides while empty state is visible.

**Open gap:** Requests should pick one primary ("Add someone" tonal, "Share my code" text) to match the Chats grammar.

**PT-5** Wide You/Requests: settings squeezed beside a stale "Select a chat" placeholder **High**

**Fixed** Panels · responsive (was R-1)

At ≥900px, Profile/Security rendered in the ~400px master column while the detail pane kept a placeholder **pixel-identical across L15/L16/L19** (verified by region diff) — 51% of a 915px viewport showing "Select a conversation" while PIN/recovery/wipe rows were cramped.

app.js:666–682 (pre-fix) · ui-audit/landscape/L15, L16, L19

**Resolution:** `.wide-solo` mode — You/Requests collapse the shell to a single content-max column; the placeholder returns only on Chats or when a chat is open.

**PT-6** Toasts repositioned for a bottom nav that still exists; 600–899px dead zone **Medium** **Fixed**

Panels · responsive (was R-2)

Wide mode lowered the toast to 1rem from the bottom on the false assumption "wide layout has no bottom nav" — the pill nav persists (overlap risk); at 600–899px the frame went full-bleed in width but kept phone height and the floating-card radius/shadow.

style.css:578–580 (pre-fix) · :68–73 (pre-fix)

**Resolution:** toast always pays `--nav-height`; the 600px breakpoint resets height/radius/shadow; nav height tokenised.

## 6 · Consistency · buttons & action hierarchy

**BT-1** Danger styling forked five ways **Medium** **Partial** Buttons · hierarchy (was C-3)

Five mechanisms paint "destructive": `.danger-fill` (filled button recoloured to error), `.danger-tonal` (tonal → error-container), `.danger-text` (raw hexes, M3-9), `.danger-outline` (named outline, actually a filled-tonal error pill), and error-coloured menu items. The polish pass removed the worst bug — a permanent inline error override on `#confirm-ok` that made *non-danger* confirmations wear error colours — and flattened the Danger Zone card-in-card. The five-way fork itself remains.

index.html:886 · style.css:812–844, 893–906, 1022, 1125–1128, 1282

**Resolved:** inline override removed (class alone decides colour); Danger Zone flattened to a labeled group with a hairline.

**Open gap:** collapse to an M3 pair — destructive primary = filled error, destructive secondary = tonal error — and rename `.danger-outline` to match what it is.

**BT-2** The confirm dialog is the only dialog whose Cancel is not a text button **Medium** **Open**

Buttons · dialog grammar

Every dialog family pairs `md-text-button` Cancel (left) with `md-filled-button` confirm (right) — edit dialog, relay offer, confirm-phrase, view-phrase, link-existing, duress steps, wipe. The highest-stakes dialog of all, `#confirm-modal`, pairs two **identical** `md-filled-tonal-button`**s**: Cancel and "Delete" are visual twins until the danger class paints one of them. Cancel and confirm therefore have equal weight in exactly the place where hierarchy matters most.

index.html:887–888 (confirm) vs :899–900 (edit) · :285–287, 495–500, 542–545, 583–585, 643–646 (the other five)

**Open gap:** make `#confirm-cancel` a `md-text-button` like every other Cancel. M3 dialog guidance: text for dismiss, filled for the confirming action.

**BT-3** The same action wears a different button style on each surface **Medium** **Open**

Buttons · consistency

"Copy" is `md-filled` on Profile, `md-text` in the phrase/device/view-phrase panels, and `md-filled` again as "Copy all" in diagnostics. "Scan QR" is `md-text` in the Chats empty state but `md-filled-tonal` on the Add screen. "Cancel" is `md-text` everywhere except confirm (`md-filled-tonal`) and the scan overlay (a bare unstyled `.scan-cancel`). Profile's account row mixes filled + tonal + text while the parallel Devices row is text + text.

index.html:439 vs :203/:496/:552 vs :997 · :381 vs :764 · :899 vs :887 vs :910 · :438–443 vs :492–494

**Open gap:** a one-page action grammar — *primary = filled, secondary = tonal or text, tertiary/dismiss = text* — applied per *role of the action*, not per screen. "Copy" should be text-or-tonal everywhere; "Scan QR" should be one of the two.

**BT-4** Accept/Decline changes grammar per request type **Medium** **Open** Buttons · consistency

Contact requests: Accept and Decline are both `md-filled-tonal-button` (decline recoloured via `.danger-tonal`). Relay requests: Approve/Decline are both `md-text-button`. Group invites: both `md-text-button`. Three request types — which sit in the same Requests tab — present the same decision with three different visual weights.

app.js:1711–1712 (requests) · :1735–1736 (relay) · :3766–3767 (group invites)

**Open gap:** one accept/decline pair for the Requests tab (recommended: Accept = filled-tonal, Decline = text) across all three row types.

**BT-5** Touch-target policy is split between 48px and a 44px "floor" **Medium** **Open** Buttons · targets

`--tap-min: 48px` is honoured by icon buttons, segments, nav items, password toggle, retry pill and toast actions. But the codebase contradicts itself: comments call 44px "the floor" for `.msg-retry` neighbours, and 44px controls remain (`.send-btn`, select-bar buttons, `.msg-menu-item`, `.msg-delete`, reaction buttons). `.mode-btn` and `.you-tab` have **no min-height at all** — padding-only, ending up ~34–37px tall.

style.css:76 (token) · :1006, 1320, 1275, 952, 1357 (44px) · :868–871, 1693–1700 (no min-height) · :964–965 ("the floor" comment)

**Open gap:** pick one policy (M3 = 48×48dp minimum, 44dp acceptable for dense visual affordances like icon-button *hit areas* expanded via pseudo-element) and enforce it on `.mode-btn`/`.you-tab` first — those are the two strips people tap repeatedly.

**BT-6** Broken icon references; ligature icons that cannot render **Medium** **Fixed** Buttons · icons

(was C-5)

The sprite had no `ic-mail` or `ic-check-circle` (referenced by the mailbox button and Select menu item); phrase Copy/Download used Material Web's boolean `LeadingIcon` ligature attribute in an app that deliberately ships no icon font.

index.html:202–203, 459, 962 · sprite block

**Resolution:** added `ic-mail`, `ic-check-circle`, `ic-visibility`, `ic-backspace`; all icons now slotted SVGs on the 24dp grid.

## 7 · Consistency · navigation

How people move: wizard stepping, chat-detail back, hardware back, overlay dismissal. Rev 1 fixed the flow-level breakages; what remains is glyph and pattern drift.

### NV-1 Hardware back corrupted wizard state; could wipe the phrase behind an open modal

High · Fixed · Navigation (was O-1)

The global back handler predated the wizard: back on the recover form yielded an empty card (targeting a hidden legacy view); back while the "Did you save your phrase?" modal was open ran `clearOnboardingSecrets()` — the modal stayed up over 24 blank words. A referenced `verify-view` no longer existed.

app.js:4820–4828 (pre-fix)

**Resolution:** wizard-aware back chain (close confirm modal → `WIZ_FLOW` step back → exit on welcome); legacy sub-box teardown for onboarding removed; phrase wiped only via explicit confirm.

### NV-2 Most action steps had no back affordance at all

High · Fixed · Navigation (was O-2)

Back/Next existed only on explainer steps; password, profile, phrase, recover and link rendered no back control — three screens, three navigation patterns (arrow icon, trailing text button, shared footer).

app.js:2832–2838 (pre-fix) · index.html:143–146, 194

**Resolution:** every step declares Back/Next in `WIZ_FLOW`; one shared footer shows Back in a consistent position (Next hidden where the primary lives in-panel); bespoke arrows removed; focus moves to each step heading with `aria-live` step counter. The hardware-back chain (scan → menus → viewer → search → sheets → dialogs → wizard step → select mode → detail → tab → exit) is now explicit and ordered.

### NV-3 Viewer opened with a back-arrow close on a modal surface

Medium · Fixed

Navigation · conventions (was M-3)

The image viewer's close control was a back arrow (modal convention is ✕); the 92% scrim ghosted the chat; the viewer also stomped bottom-nav visibility unconditionally (wide-layout desync).

app.js:3913–3927 (pre-fix) · style.css:1104 · index.html:943

**Resolution:** ✕ close, opaque backdrop, nav visibility restored via the same wide/narrow rule as `showDetail`; zoom % pill and one-time "Double-tap to zoom" hint added.

### NV-4 Dismissal still forks into three glyphs

Low · Open · Navigation · conventions

Back-ish navigation uses `#ic-back` arrow (`#back-btn`, search close); modal dismissal uses `#ic-close` ✕ (image viewer, account-fs); the diagnostics sheet closes with a raw `×` text glyph that is neither of them and inherits no icon-grid metrics.

index.html:736 (arrow) · :946, :917 (✕) · :995 (raw ×)

**Open gap:** one rule — *arrow = navigate back, ✕ = dismiss overlay* — and slot `#ic-close` into the diagnostics sheet.

## 8 · Consistency · selection

How people choose: long-press menus, multi-select, segmented toggles, swatches, pickers. This is the least consistent interaction family in the app — and the one where hand-rolled components (M3-12) cost the most.

### SL-1 Keyboard/switch users could not use menus or dialogs

High · Partial · Selection · accessibility

(was A-1)

`msg-menu`/`react-bar` (`role="menu"`) opened without focus or key handling; no modal trapped Tab; focus was never restored to the trigger; wizard step changes dropped focus to `<body>` with no announcement.

app.js:299–392, 2865–2878, 458–468 (pre-fix)

**Resolved:** shared Tab trap + focus-restore on the confirm/edit dialogs; wizard focuses each step heading and announces "Step X of 8" via `aria-live`; viewer focuses its close control.

**Open gap:** menus still have no keyboard interaction (see SL-3); this is what the deferred `md-menu` migration buys.

### SL-2 Multi-select carries no ARIA, and its state class collides with long-press

Medium · Open

Selection · semantics

`selectMode` adds `.selecting` rows and injects `.select-check` circles, but rows carry no `aria-selected`/`aria-multiselectable`, the contextual bar announces nothing, and the same `.selected` class is used for the transient long-press highlight (8% primary tint on the bubble) as for a checked item — so "highlighted because pressed" and "checked for action" are visually and semantically the same thing. `#select-delete` also fakes disabled via opacity in one path (fixed in Rev 1 to a real `disabled`, but the styling contract remains ad hoc).

app.js:4980–5043 (selectMode) · :4662–4668 (long-press highlight) · style.css:1285–1301

**Open gap:** `role="listbox"`/`aria-multiselectable` on the message list during select mode (or `aria-pressed` per row), an accessible name on the select bar ("3 selected"), and distinct visuals for pressed vs checked.

### SL-3 Overflow menus have no menu roles; the one menu that has them has no keys

Medium · Open · Selection · menus

The chat and group overflow menus (`#chat-menu`/`#group-menu`) are plain `.menu` divs of bare buttons with **no roles at all**. The long-press message menu does declare `role="menu"`/`menuitem` — but on open, focus is never moved into it, arrow keys do nothing, and only outside-tap/Escape dismiss it. The forward picker declares `role="listbox"` whose rows are plain buttons with no `role="option"`.

index.html:739–747 (no roles) · :962–973 (roles, no keys) · :979 + app.js:4896–4908 (listbox without options) · app.js:4654–4753 (anchoring, no focus)

**Open gap:** the fastest complete fix is `md-menu` (keyboard, focus, dismissal for free); the minimum fix is one `openMenu(el, trigger)` helper — focus first item, ↑/↓/Home/End/Escape, restore focus — used by all four menus.

### SL-4 Single-selection is class-only everywhere except one picker

Medium · Open · Selection · semantics

Theme `.mode-btn`, You tabs, `.seg` toggles and accent swatches all express "chosen" purely via a CSS class. The only true single-select semantics in the app is the relay picker: `role="radiogroup"` with `role="radio"`/`aria-checked` items. So screen-reader users can operate exactly one of the app's five "choose one" widgets.

app.js:157–164, 881–890, 2609, 3932 (class toggles) · :775 (aria-selected on you-tab) · :4014–4037 (the correct pattern)

**Open gap:** apply the relay picker's pattern to the other four, or use native radio semantics / `md-radio` where the widget is a radio group (wipe threshold, theme mode).

### SL-5 Truthful-affordance defects

Medium · Fixed · Selection · affordances (was A-3)

Fullscreen QR hint said "Tap anywhere to close" while only the backdrop closed — and rendered on-surface-variant gray over an 85% scrim (~2:1); the PIN pad's backspace wore the same glyph as screen-back; `#select-delete` faked disabled via opacity.

index.html:919, 915 · app.js:544, 4702 (pre-fix)

**Resolution:** honest "Tap the background to close" in forced white + ✕ close; backspace glyph added; real `disabled` attribute.

## 9 · Flow-level findings (preserved from Rev 1)

Kept for traceability. These are journey-level defects rather than design-language ones; the design-language consequences of each are captured above.

**FL-1** Most consequential chat text was the smallest — and not a control **High** **Fixed** Hierarchy

(was T-2)

"Not delivered — couldn't send · tap to retry" rendered at 0.7rem (~11px) with a ~15px hit target (⅓ of the 44px floor) while the failed image was a 260px dimmed square whose tap opened the full viewer.

style.css:872 · app.js:1704–1714, 3828

**Resolution:** failure cause moves to a label-size error line; **Tap to retry** is a ≥48px error-container pill with refresh icon; group chats gain the same states (previously none).

**FL-2** Wizard PIN step validated nothing; empty submit fired a raw server error **High** **Fixed**

UX practice (was O-3)

Label promised "6+ digits"; only equality was checked. But the deeper defect was that the step **could not succeed for any user, ever**: `POST /api/security` is `s.auth()`-gated on a vault session token, and no session exists anywhere in the wizard — `setSessionToken()` runs only in `doUnlock()`, reached from the *"Start messaging"* button two steps later. `pinEnable` needs the live unlocked identity (it reads `MnemonicString()`/`PassphraseString()`), so no pre-unlock session could ever have served it. Reproduced on a host instance: `POST /api/create` returns `{account_number, recovery_phrase}` with **no** `session_token`, and the pin-enable call answers `401` with the plain-text body `unauthorized` — not JSON, so `body.error` was undefined and the UI always fell back to its own generic string (walkthrough `13c`: "Could not turn on PIN unlock.").

app.js:2957–2968 (pre-fix) · ui-audit/13c, L13c

**Resolution:** client validates `/^\d{6,}$/` + match with inline field-adjacent errors; the step then *holds* the PIN (`wizPendingPin`) and makes no network call at all, and `wiz-finish` applies it immediately after `doUnlock()` mints the session — the first moment the endpoint can succeed. Deferring was chosen over minting a session at the PIN step: a live token mid-wizard makes `refresh()` take its `sessionToken && unlocked` branch and drop the user into the chat view on the next WebView rebuild, and it would start the onion service three steps earlier, moving publish timing for no user-visible gain. A genuine failure now surfaces as a snackbar naming You›Security rather than silently leaving the PIN off. **No backend change was required** — the earlier follow-up (widen `POST /api/security` to a "pre-unlock onboarding session") was misdirected: there is no onboarding session to widen to, and inventing one would move an auth boundary for one wizard step. **Verified** by driving the shipped wizard over CDP against a host instance: pre-fix the same script reproduces frame `13c` exactly (stuck on step `pin`, banner visible, `pin_enabled:false`); post-fix the PIN step makes zero requests, advances to `done`, and `GET /api/security` reports `pin_enabled:true`. "Not now" still ends at `pin_enabled:false`.

### FL-3 Three validation philosophies in one wizard; red ✗ mid-keystroke

Medium · Partial

Consistency (was O-4)

Create/recover gated the button until valid; link/PIN validated on submit into a detached paragraph. Requirement rules and the match icon went red while the user was still typing.

app.js:2670–2674, 2703–2714, 2992–3019

**Resolution:** PIN and profile now validate with precise inline errors; progressive green ✓ retained while typing. Full blur-then-error conversion of the requirements checklist deferred (§12).

### FL-4 Notify explainer lacked its CTA; terminology drifted across three names

Medium · Fixed

Copy (was O-5)

The notifications step told users to go enable notifications manually; the same concept was taught as "account number", done-step as "friend code", link form as "pairing code".

index.html:214, 236, 261, 174

**Resolution:** "Turn on notifications" button fires the permission request (Android: honest pointer to system settings); **"friend code"** is the single user-facing name; "pairing code" reserved for device-link with clarifying copy.

### FL-5 "Remove account" password gate was a silent no-op on Android

High · Fixed · UX practice (was S-1)

Confirmation was followed by `window.prompt()` — unsupported by Android WebView (no `onJsPrompt`). The reddest button in the app showed no dialog, got `null`, and aborted with zero feedback.

app.js:2496 (pre-fix) · MainActivity.kt:341–377

**Resolution:** replaced with the app's own `editDialog({passwordFirst:true})` — styled field, show/hide, error surface.

### FL-6 PIN setup: silent mismatch discard; API errors rendered as editable fields

High · Fixed

UX practice (was S-2)

A PIN mismatch closed the dialog and dropped the input with no message; a server failure re-opened the dialog with the error text as a text-field *label* — an editable "field" reading "Could not turn on PIN unlock."

app.js:2295–2315 (pre-fix)

**Resolution:** dialog stays open across failures; static `role="alert"` error above the actions; values preserved; confirm verb "Turn on PIN".

**FL-7** "Log out" actually just locked; settings writes failed silently **Medium** **Fixed** Copy (was S-3)

The label sat next to genuine destruction while only locking the session; name/mailbox/group-backup/primary-device writes `return`ed on failure with no feedback.

index.html:650 · app.js:3173, 3215, 3233, 3298

**Resolution:** renamed **"Lock Molepipe"**; all four failure paths surface the server error via snackbar; wrong-PIN countdown split into its own status line; PIN keypad gains a real backspace glyph, per-key haptics and a "Keypad shuffled" notice.

**FL-8** No delivered state existed — success was the disappearance of "Sending…" **High** **Fixed**

UX practice (was M-1)

Outgoing messages rendered "Sending…" or a red failure; on ack the status element vanished. Pixel analysis confirmed the capture set's "delivered" frame was byte-identical to the failed frame. Group chats showed no delivery UI at all.

app.js:1704–1714 (pre-fix) · ui-audit/54 vs 54b (MD5-identical but clock)

**Resolution:** explicit terminal state — check icon + "Delivered" (label role, on-surface-variant) for 1:1 *and* groups; red reserved for failure. Wire protocol: `delivery` is `pending`/`failed`/cleared — cleared reads as delivered.

**FL-9** Friend lookup: ~80s cold Tor dial behind a dead button **High** **Fixed** UX practice (was M-2)

"Looking up…" disabled the button with no progress, cancel, hint or timeout; `api()` had no AbortController, so a hung fetch left the button stuck forever.

app.js:282–290, 3373–3380 (pre-fix)

**Resolution:** `api()` carries a 95s abort timeout returning an honest "network is very slow" error; lookup shows an indeterminate `md-linear-progress` + "can take a minute on a cold start" hint; try/catch so network faults can't strand the button.

**FL-10** Image send flaky (intermittent, not orientation-specific) **Medium** **Backend** UX practice (was M-4)

First portrait pass: 0/2 images delivered while text succeeded; landscape pass and portrait retest delivered 3/3 with no code changes. Treated as intermittent `/api/messages/image` delivery failure.

ui-audit/54b, 62, 64 vs landscape L35–L36 and portrait-retest/02–03

**Follow-up:** backend investigation of the image outbox path; UI retry pill (FL-1) now gives users a first-class recovery action.

## 10 · Strengths preserved

- **Complete M3 colour foundation** — 36 system roles with a full surface ladder, seven accent schemes, and runtime Material You palettes bridged from Android; surfaces consistently draw from `surface-container-*` roles.

- **Honest Tor status system** — four token-coloured states, a split "Send ✓ / Receive ⟳" pill that refuses to say Online before the descriptor publishes, determinate bootstrap bar, stall detection; the pill remains tappable into Diagnostics with an honest accessible name.
- **Lock-screen security UX** — no fixed-length PIN placeholders, scramble-on-failure, explicit attempt countdown, forgot-PIN escape always one tap away, consequence-first destructive disclosure copy.
- **One confirmation architecture** — every consequential action routes through `confirmDialog` with honest titles and consequence-named buttons.
- **Explicit hardware-back chain** — one ordered handler (overlays → menus → viewer → dialogs → wizard → select mode → detail → tab → exit) with Esc delegating to it.
- **Text-safety discipline** — user-controlled strings only ever reach `textContent`/`createTextNode`; the three `innerHTML` template sites are static-only and documented.
- **Long-press implementation** — 500ms with cancel slop, click-swallow and haptic; the image viewer's pinch/double-tap separation is above average.
- **Numbered phrase grid** — CSS-counter ordinals keep `textContent` as the bare 24 words, which is what lets secret-wiping work reliably.
- **Two-pane master-detail** — divider at the 400px minimum, chat pane exempt from the form-width cap, bubbles capped at a readable `min(82%, 620px)` measure.

## 11 · M3 polish pass — what changed

Implemented across `style.css`, `index.html`, `app.js` (embedded via `go:embed`, shipped through the gomobile AAR). All High findings resolved; verification build deployed to `emulator-5554`.

| Layer | Changes |
|---|---|
| **Tokens** | M3-mapped type roles (22 sizes → 8), spacing scale, elevation ladder, scrims, motion vars, layout constants, z-scale, semantic status colours — 114 codemodded sites |
| **Colour** | Donate card onto `tertiary-container`; status colours consolidated; inline styles 34 → 4 |
| **Dialogs** | One M3 spec + roles/aria, focus trap & restore, action-specific verbs, error-as-static-text, narrow modifier |
| **Feedback** | M3 snackbar queue with actions; 7 channels unified; success out of alert regions; silent failures surfaced |
| **Messaging** | Delivered ✓ (1:1 + groups), 48px retry pill, lookup progress + 95s API timeout |
| **Viewer** | ×, opaque backdrop, zoom %, first-use hint, Not-sent chip, actions gated, nav desync fixed |
| **Wizard** | Back everywhere + safe hardware back, PIN validation, notify CTA, friend-code terminology, focus/aria |
| **Settings/lock** | editDialog for erase-account, PIN dialog error handling, Lock Molepipe, icon fixes, keypad glyph/haptics/shuffle notice, split PIN error |
| **Responsive** | wide-solo You/Requests, toast offset, 600px shell reset |
| **A11y** | Global focus ring, complete reduced-motion, 48dp targets on key controls, real disabled, truthful hints, 4 sprite icon fixes |

**Live regression fix during QA:** the polish pass initially made the Tor pill passive (audit note: it looked like a chip but opened the log). User testing flagged the pill as "the log button" — tap-to-open was restored with the honest label *"Tor status — tap for diagnostics"* and verified on the emulator (pill → Diagnostics sheet with live snapshot + Tor log).

## 12 · Open gaps & M3 migration roadmap

The 16 open gaps from this consistency pass, in the order that buys the most consistency per unit of risk.

| Priority | Item | Gaps closed | Notes |
|---|---|---|---|
| **1** | One action grammar for buttons (per role, not per screen) | BT-2, BT-3, BT-4, BT-1 | Cheapest high-visibility win: confirm-cancel → text button; one Copy/Scan-QR/Cancel style; one Accept/Decline pair; danger → filled/tonal only |
| **2** | One selected-state rule + one switch mechanic | PT-1, PT-3 | Segmented = primary-container, tabs/nav = secondary-container; `showPane()` primitive for hidden vs class swaps |
| **3** | Finish the tab/nav ARIA patterns | PT-2, SL-4 | `aria-current` on nav; full ARIA tabs (controls, roving tabindex, arrows) on You tabs; radiogroup semantics on wipe/theme |
| **4** | `md-menu` migration (or one `openMenu()` helper) | SL-1, SL-3 | Gains keyboard interaction, focus management and dismissal for all four menus; also removes the forward-picker listbox lie |
| **5** | Select-mode semantics | SL-2 | `aria-selected`/listbox or `aria-pressed`; accessible count; pressed vs checked visuals |
| **6** | Target-size policy enforcement | BT-5 | `.mode-btn`/`.you-tab` min-height first; resolve 44 vs 48 in the code comments |
| **7** | Motion + state-layer tokens actually adopted | M3-5, M3-13 | Route surfaces through `--ease-emphasized`+durations (or delete them); `md-ripple`/`md-focus-ring` on hand-rolled controls |
| **8** | Token hygiene | M3-6, M3-9, NV-4 | Spacing codemod or token removal; `.danger-text` onto error roles; ✕ into the diagnostics sheet |
| **9** | `md-dialog` pilot on `#confirm-modal`, then the rest | M3-10, M3-12 | Div-dialogs are spec-unified and safe today; this is behaviour parity, not a fix |

| Also deferred | Rationale / next step |
|---|---|
| Blur-then-error on password requirements | Green-✓ progression kept live; red ✗ deferral pending a pass over `updatePwRequirements` (FL-3) |
| Full-row tap on switch settings rows | Label wrap → switch; rows currently toggle only via the switch widget |
| Navigation rail at ≥1200px | Product chose to keep the M3 navigation bar; rail remains an option for desktop-first use |
| List/chat skeletons | Only image bubbles shimmer today; the 2.5s poll rebuild can flash empty→content on first paint |
| ~~Backend: wizard PIN enable~~ — **done, client-side** | Resolved without touching the backend: the wizard holds the chosen PIN and applies it right after the unlock that ends setup. See FL-2 — the "pre-unlock session" this row asked for does not exist, so there was nothing for `POST /api/security` to accept |
| **Backend:** image send flakiness | Intermittent `/api/messages/image` delivery failure (FL-10); UI retry exists |
| Dark-mode capture pass | Entire audit set is light-theme; dark mode verified only via token analysis (donate card fix is token-driven) |
| Screenshot re-capture | Re-run the CDP workflow; repair mislabeled frames (`54`, `61`, `L41`) and add new states (delivered ✓, retry pill, snackbar, wide You-tab, dark) |

## 13 · Verification

- **Static (Rev 1):** `node --check app.js`; HTML well-formedness (balanced stack, zero errors); CSS brace balance (551/551); every `$("id").addEventListener` target cross-checked against `index.html` — zero dangling references.
- **Deployed:** gomobile AAR rebuilt (UI is `go:embed`ded) → `assembleDebug` → installed on `emulator-5554`; unlocked via the scrambled PIN pad; tor-pill → Diagnostics verified end-to-end.
- **Source re-audit (Rev 2):** every claim in §3–§8 re-verified against current `file:line` locations, including a full census of `md-*` tag usage vs the vendored bundle and a zero-reference scan for the motion/spacing tokens.
- **Live wizard replay (FL-2):** the onboarding flow is now exercisable without touching a device account — `cmd/server -data-dir <tmp> -no-tor -open=false` serves the same `go:embed`ded UI with no identity on disk, so the wizard renders and can be driven over CDP with headless Chromium. Used to confirm FL-2 both ways: the identical script reproduces frame `13c` against a binary built from the pre-fix `app.js`, and passes against the fix. This is the cheapest route to the screenshot re-capture still pending below, and the only one that can reach the pre-account steps at all.
- **Pending:** full portrait + landscape screenshot re-capture and pixel-diff against this report's baseline (`ui-audit/`), including the first dark-theme pass.

## Appendix A · Screenshot index (selected)

| Frame(s) | State | Related finding |
|---|---|---|
| `05-06b` | Password validation states | FL-3, M3-8 |
| `13c`, `L13c` | Wizard PIN enable error | FL-2 |
| `14` | Done step (donate card) | M3-3 |
| `15`, `L15` | First unlock / two-pane placeholder | PT-4, PT-5 |
| `19`, `22-23` | Security pane, PIN setup, PIN on | FL-5–FL-7, BT-1 |
| `20` | Fullscreen QR ("tap anywhere") | SL-5 |
| `27-28` | Lookup loading → result | FL-9 |
| `33-36`, `L34` | Sending → delivered states | FL-8 |
| `38` | Long-press menu + reactions | M3-1, SL-1, SL-3 |
| `39` | Requests tab empty | PT-4, BT-4 |
| `40-41` | PIN pad, wrong PIN | FL-7, SL-5 |
| `45` | Remove-account modal | FL-5, BT-2 |
| `54b`, `62`, `64` | Image not delivered + retry | FL-1, FL-10 |
| `55-61`, `L37-L41` | Viewer, zoom, actions bar | NV-3 |
| `L16`, `L19` | Wide You/Security panes | PT-5 |

## Appendix B · Finding ID map (Rev 1 → Rev 2)

| Rev 1 | Rev 2 | Rev 1 | Rev 2 |
|---|---|---|---|
| DS-1 | M3-1 | O-1 | NV-1 |
| DS-2 | M3-2 | O-2 | NV-2 |
| DS-3 | M3-4 | O-3 | FL-2 |
| DS-4 | M3-3 | O-4 | FL-3 |
| C-1 | M3-11 | O-5 | FL-4 |
| C-2 | M3-10 | S-1 | FL-5 |
| C-3 | BT-1 | S-2 | FL-6 |
| C-4 | PT-4 | S-3 | FL-7 |
| C-5 | BT-6 | M-1 | FL-8 |
| T-1 | M3-7 | M-2 | FL-9 |
| T-2 | FL-1 | M-3 | NV-3 |
| T-3 | M3-8 | M-4 | FL-10 |
| A-1 | SL-1 | R-1 | PT-5 |
| A-2 | BT-5 (targets) · M3-13 (focus/motion) | R-2 | PT-6 |
| A-3 | SL-5 | — | new: M3-5, M3-6, M3-9, M3-12, PT-1–PT-3, BT-2–BT-4, NV-4, SL-2–SL-4 |

Generated from the interactive audit session. Source captures and walkthrough: `ui-audit/README.md` and sibling PNGs. Report file: `ui-audit/Molepipe-UI-UX-Audit.html` (regenerate the PDF with headless Chromium).